Evaluating AI Agents

Improving your LLM-as-a-Judge





DeepLearning.AI

In this lesson:

- How to tailor your LLM evaluators to your agent

Improving your Evaluators

Thinking back: Why would you use two evaluators to evaluate the same thing, like you did with function calling?

Component of your agent	**Example test case**	**Experiment**	**Evaluations**
Router	{'input': 'Which stores had the best sales performance in 2021', 'expected_output': 'database_lookup'}	Testing different tool description strings	1. Code-based comparison against ground truth 2. Function calling using LLM-as-a-Judge*

Improving your Evaluators

Thinking back: Why would you use two evaluators to evaluate the same thing, like you did with function calling?

To see how accurate the LLM-as-a-Judge eval is compared to the 100% accurate, code-based comparison against ground truth

Remember, LLM-as-a-Judge is not a 100% accurate method!

Using Experiments to Improve your LLM-as-a-Judge

You can use experiments to test your LLM Judges:

LLM Judge	Example test case	Experiment	Evaluations
Function calling LLM Judge	{'input': '{'input': 'Which stores had the best sales performance in 2021', 'expected_output': 'database_lookup'}', 'expected_output': 'correct'}	Testing different LLM-as-a-Judge prompts	1. Code-based comparison against ground truth

arize

DeepLearning.AI

Using Experiments to Improve your LLM-as-a-Judge

You can use experiments to test your LLM Judges:

LLM Judge	Example test case	Experiment	Evaluations
Function calling LLM Judge	{'input': '{'input': 'Which stores had the best sales performance in 2021', 'expected_output': 'database_lookup'} }', 'expected_output': 'correct'}	Testing different LLM-as-a-Judge prompts	1. Code-based comparison against ground truth
Analysis Clarity LLM Judge	{'input': 'In 2021, the stores that performed the best based on total sales value were:....', 'expected_output': 'The analysis is clear because....'}	Testing different models to judge the analysis	How do we evaluate this output? What if the output was "The analysis is easy to understand…"

Using Experiments to Improve your LLM-as-a-Judge

You can use experiments to test your LLM Judges:

LLM Judge	Example test case	Experiment	Evaluations
Function calling LLM Judge	{'input': '{'input': 'Which stores had the best sales performance in 2021', 'expected_output': 'database_lookup'} }', 'expected_output': 'correct'}	Testing different LLM-as-a-Judge prompts	1. Code-based comparison against ground truth
Analysis Clarity LLM Judge	{'input': 'In 2021, the stores that performed the best based on total sales value were:....', 'expected_output': 'The analysis is clear because....'}	Testing different models to judge the analysis	1. Semantic similarity

arize

DeepLearning.AI

What you've learned:

How to measure and improve your LLM-as-a-Judge evals